

Recap Coverage Index

This file maps the revision checklist to the weekly notes. The main notes stay at the repo root so existing asset paths such as `assets/bin-to-dec.png` keep working.

Operating Systems and Computer Architecture

Topic	Covered in
IEEE 754 and floating point IEEE	Week 1
2's complement	Week 1
2's complement addition	Week 1
Add/subtract in binary	Week 1
Number-system conversion	Week 1
Finite state automata	Week 1
Introduction to encryption	Week 1
Moore's Law	Week 2
Digital circuits	Week 2
MUX, DeMUX, flip-flops	Week 2
MIPS	Week 2
Clock rate	Week 2
Registers and CPU registers	Week 2
RISC and CISC	Week 2
Memory: cache, SRAM, DRAM	Week 2
NAND, NOR, universal gates	Week 2
Linux commands	Week 3
Hardware interrupts	Week 3
Open source OSes	Week 3
Kernel and kernel mode	Week 3
Monolithic and microkernel structures	Week 3
System calls	Week 3
Processes, parents, children, <code>fork()</code>	Week 3
PCB and PCB sections	Week 3
Text, data, heap, stack	Week 3
Bash <code>/bin/bash</code>	Week 3

Topic	Covered in
Virtual memory	Week 3

Networking

Topic	Covered in
Protocols and RFCs	Week 4
IP suite	Week 4
TCP/IP five-layer model	Week 4
Segments, datagrams, packets, frames, bits	Week 4
Application layer and HTTP	Week 4
HTTP status codes	Week 4
Telnet	Week 4
Client/server and asynchronous communication	Week 4
Sockets	Week 4
Physical media: wired and wireless	Week 4
TCP and UDP	Week 4
Network layer	Week 5
IP addressing and subnet masking	Week 5
ARP	Week 5
Link layer and CSMA	Week 5
CRC	Week 5
FTP, FTPS, SFTP, SSH	Week 5
SSL/TLS layer and uses	Week 5
HTTP security	Week 5
DHCP and DORA	Week 5
DNS and TLDs	Week 5
Netkit and Kathara labs	Week 5

Acronyms

Acronym	Meaning
CSMA	Carrier Sense Multiple Access
CRC	Cyclic Redundancy Check
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
RFC	Request for Comments
SSH	Secure Shell
SSL	Secure Sockets Layer
TLS	Transport Layer Security
TLD	Top-Level Domain